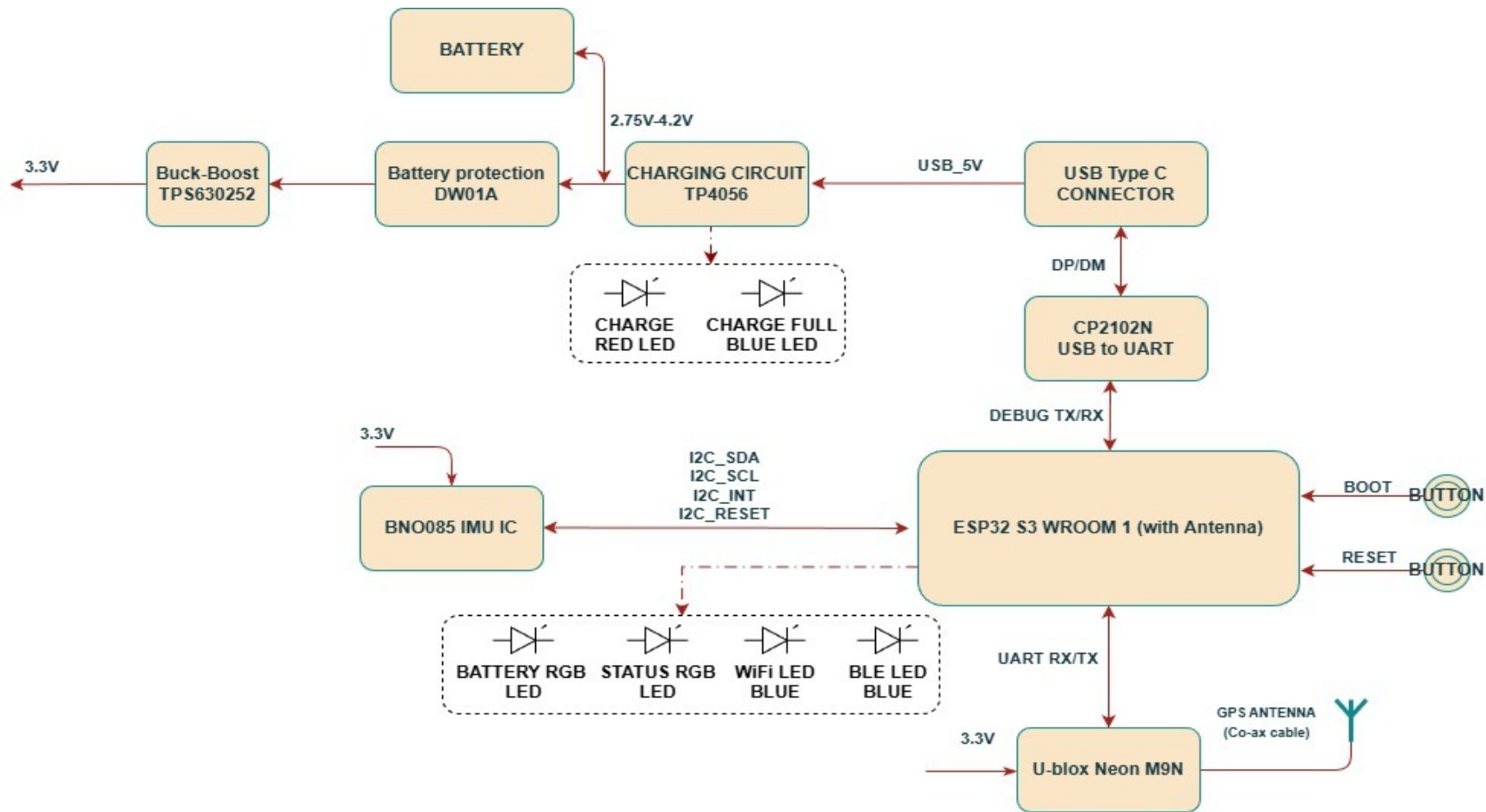


SYSTEM BLOCK DIAGRAM



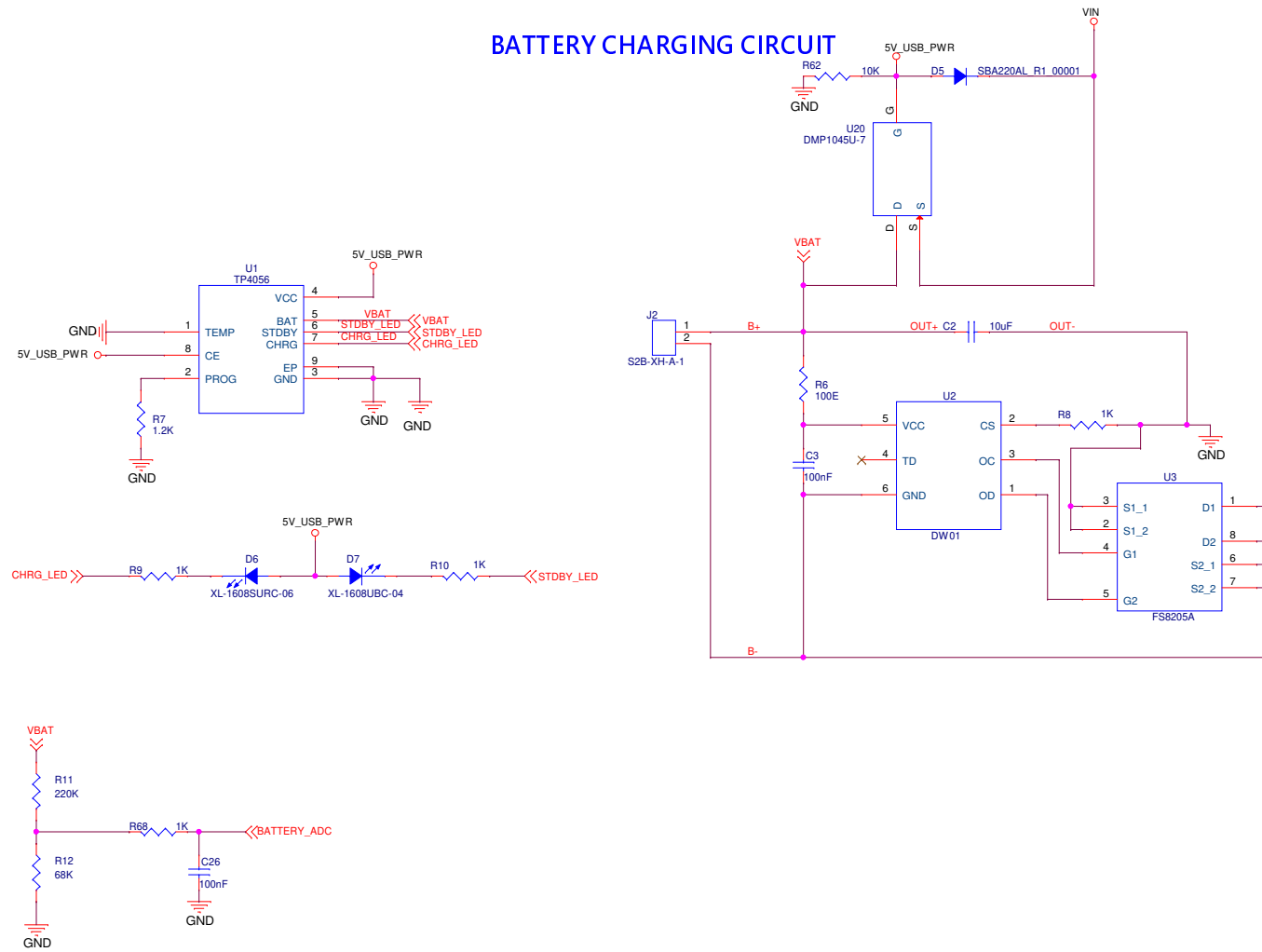
Title			CTR-GEN 3 HUB	
Size	Document Number		Rev	
A4	SYSTEM BLOCK DIAGRAM		A	
Date:	Friday, November 07, 2025	Sheet	1	of 9

The schematic diagram illustrates the internal wiring of a USB Type-C connector (J1) for the CTR-GEN 3 HUB. The connector pins are connected to various components as follows:

- SH1_G (S1) and SH2_G (S2):** Connected to a 0(1%) resistor (R1) leading to GND.
- A1_B12 (A1) and B12:** Connected to the VUSB_IN line.
- A4_VBUS (A4) and B9:** Connected to the VUSB_IN line.
- B8_SBU2 (B8) and A5_CC1 (A5):** Connected to GND.
- B7_DN2 (B7) and A6_DP1 (A6):** Connected to the DEBUG_USB_DM line.
- A7_DN1 (A7) and B6_DP2 (B6):** Connected to the DEBUG_USB_DP line.
- A8_SBU1 (A8) and B5_CC2 (B5):** Connected to GND.
- A9_VBUS (A9) and B4_A9:** Connected to the VUSB_IN line.
- B1_A12 (B1) and A12:** Connected to the VUSB_IN line.
- SH3_G (S3) and SH4_G (S4):** Connected to GND.
- R4 and R5:** 5.1K(1%) resistors connected between the VUSB_IN line and GND.
- D1:** A B5819W diode connected between the VUSB_IN line and the 5V_USB_PWR line.
- R2 and R3:** 0(1%) resistors connected between the DEBUG_USB_DM and DEBUG_USB_DP lines and the DEBUG_USB_D- and DEBUG_USB_D+ lines.
- D2, D3, and D4:** ESDB5V0WT diodes connected between the VUSB_IN line and GND.
- C1:** A 22uF/6.3V capacitor connected between the 5V_USB_PWR line and GND.

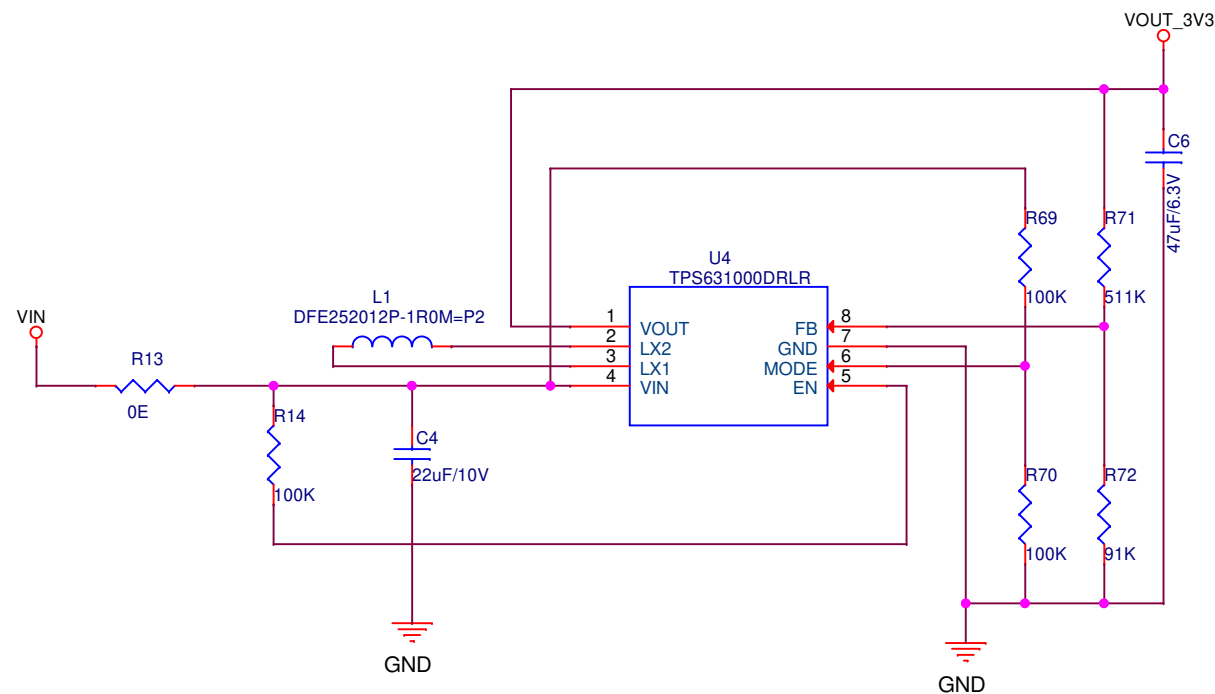
The diagram also shows the power supply (5V_USB_PWR) and ground (GND) connections for the connector.

BATTERY CHARGING CIRCUIT



Title			CTR-GEN 3 HUB		
Size	Document Number				Rev
B	BATTERY CHARGING CIRCUIT				A
Date:	Friday, November 07, 2025				Sheet 3 of 9

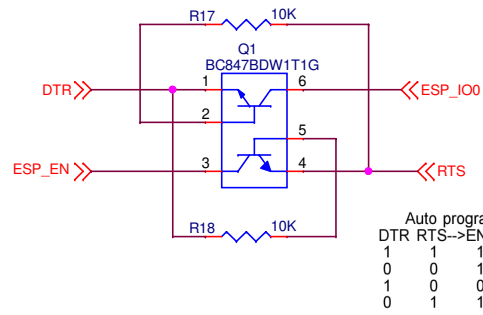
3.3V_BUCK BOOST CIRCUIT



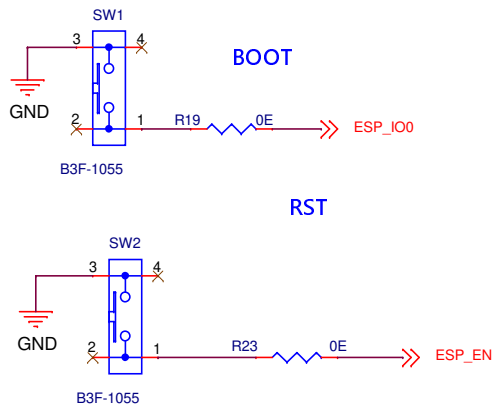
Title		
CTR-GEN 3 HUB		
Size A	Document Number 3.3V_BUCK BOOST CIRCUIT	Rev A
Date:	Friday, November 07, 2025	Sheet 4 of 9

CP2102N_USB_TO_UART_CONVERT

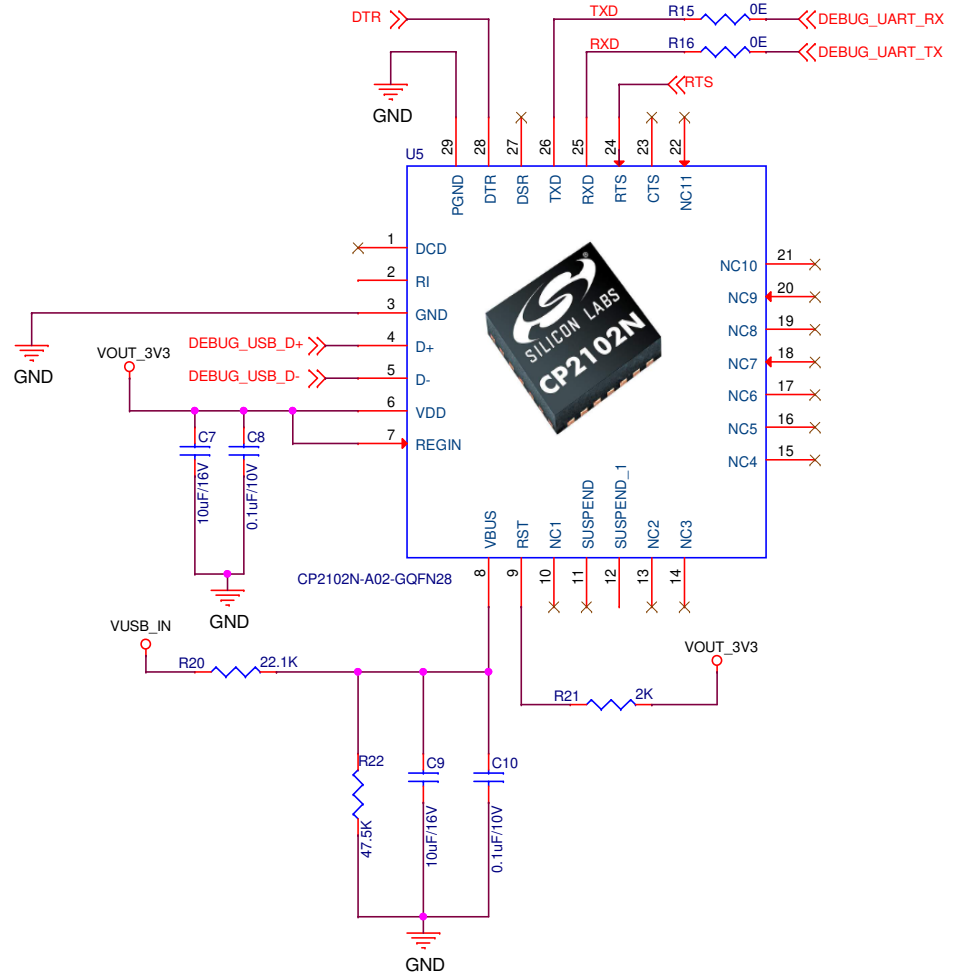
AUTO PROGRAMING CONTROL&ISOLATION



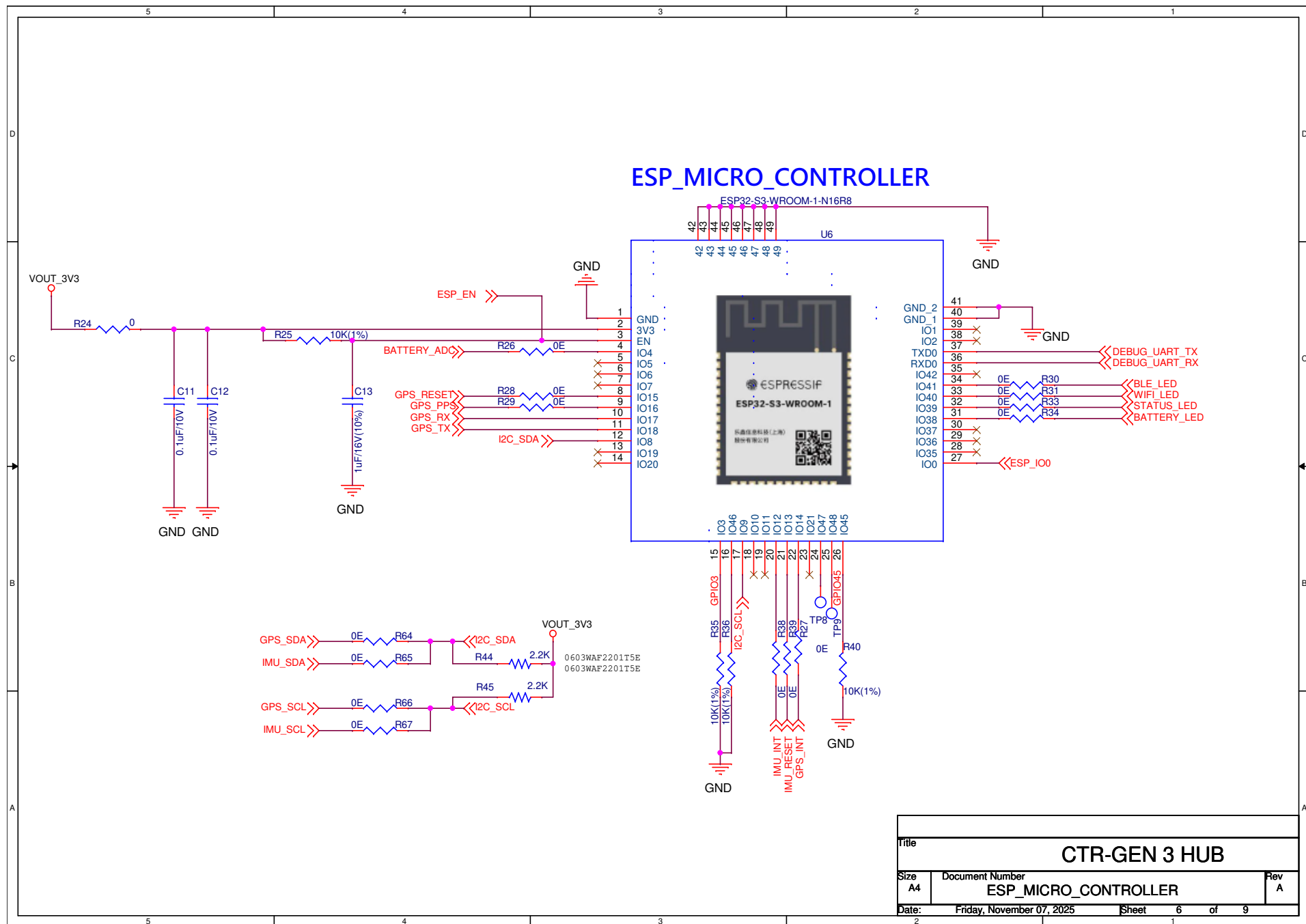
BUTTONS



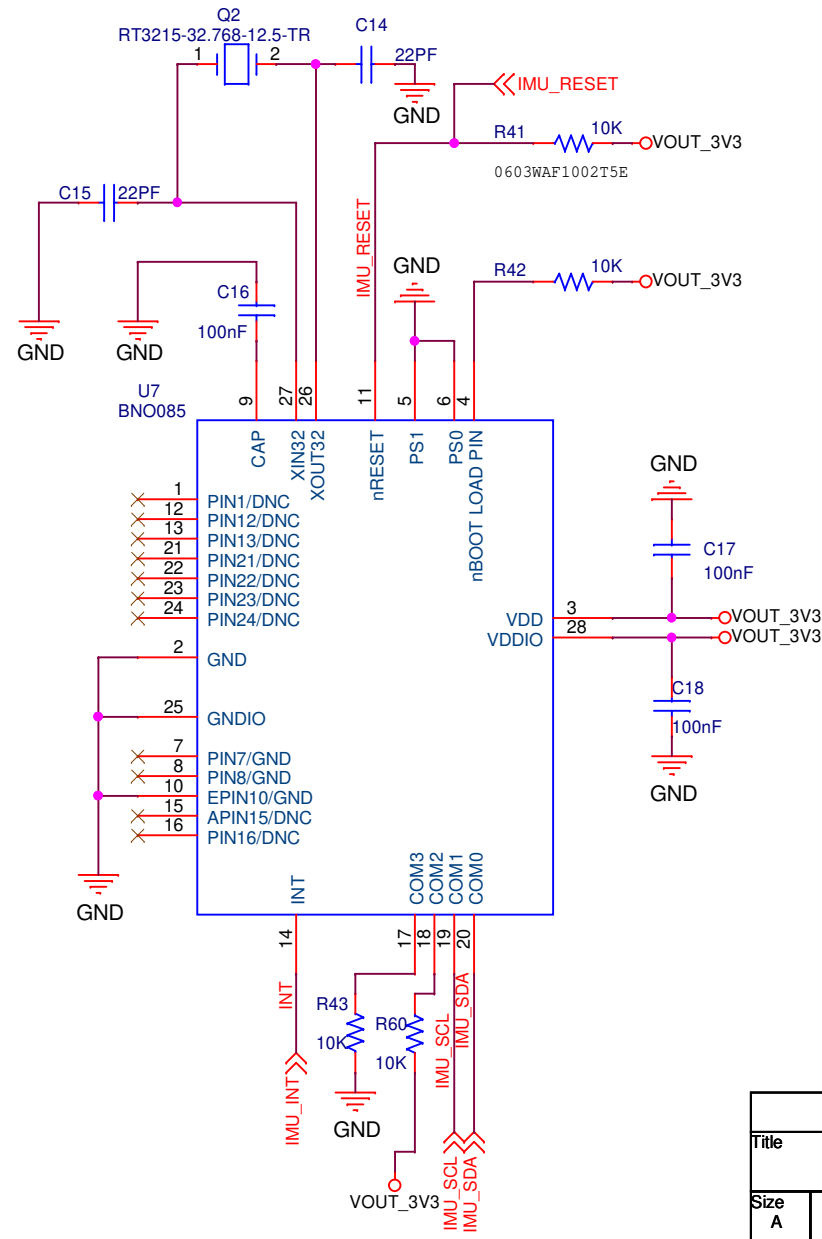
USB TO UART BRIDGE



Title		
CTR-GEN 3 HUB		
Size	Document Number	Rev
A4	CP2102N_USB_TO_UART_CONVERT	A
Date:	Friday, November 07, 2025	Sheet 5 of 9



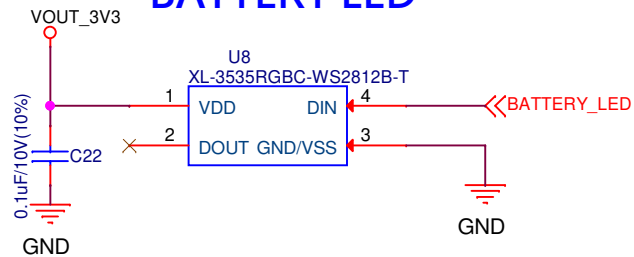
BNO085_IMU_CIRCUIT



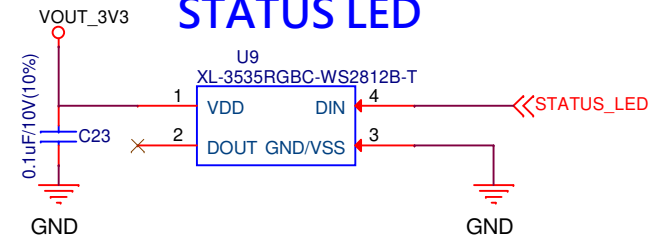
Title		
CTR-GEN 3 HUB		
Size A	Document Number	Rev A
BNO085_IMU_CIRCUIT		
Date:	Friday, November 07, 2025	Sheet 7 of 9

LEDS&TEST POINTS

BATTERY LED



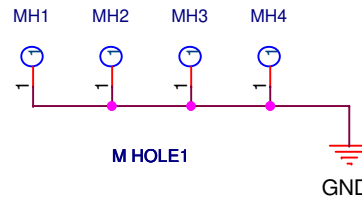
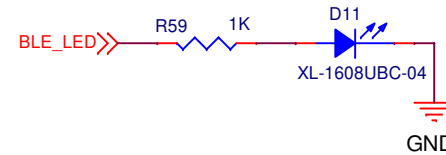
STATUS LED



WIFI LED

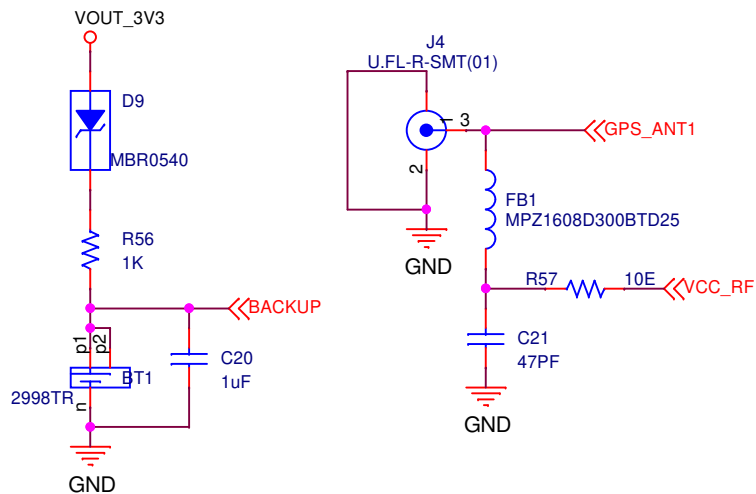
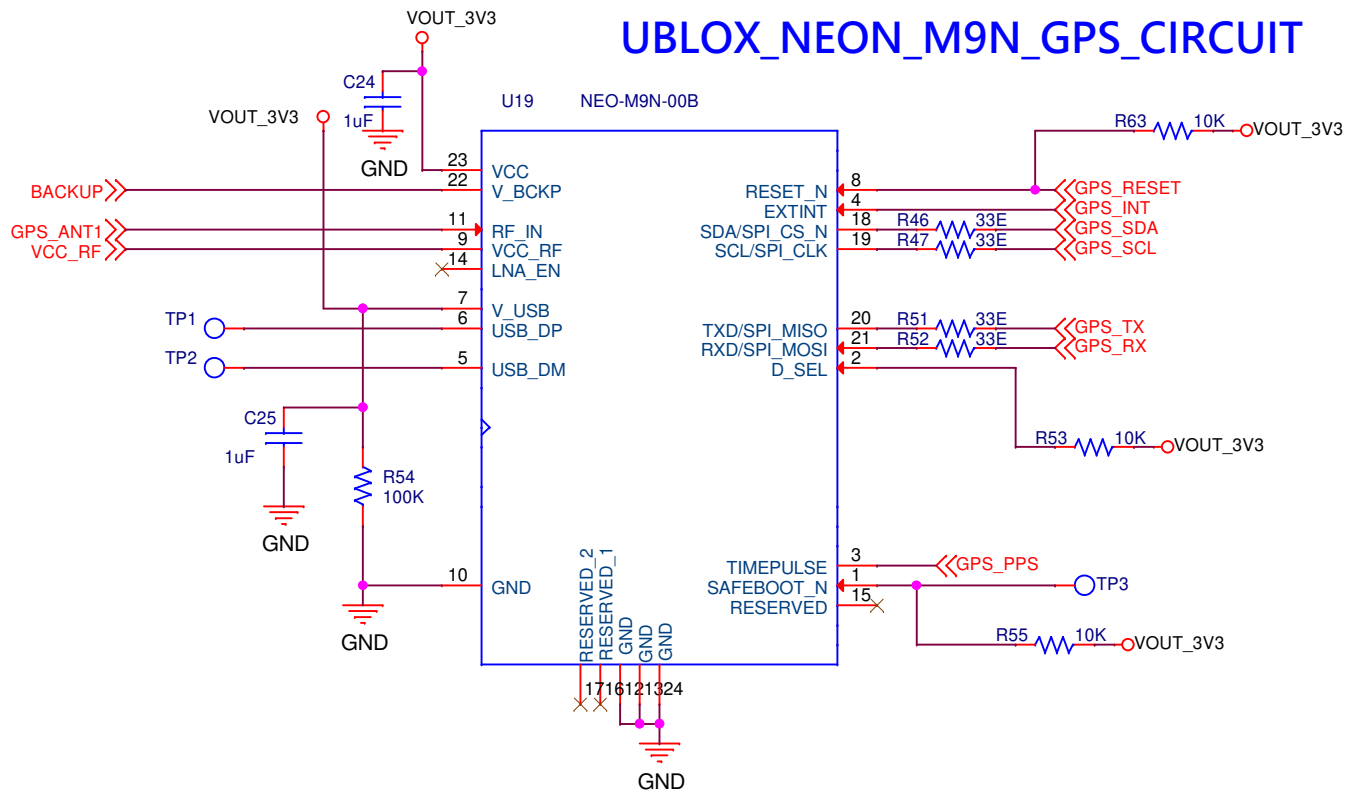


BLE LED



Title		
CTR-GEN 3 HUB		
Size A	Document Number LEDS&TEST POINTS	Rev A
Date:	Friday, November 07, 2025	Sheet 8 of 9

UBLOX_NEON_M9N_GPS_CIRCUIT



Title		
CTR-GEN 3 HUB		
Size A	Document Number UBLOX_NEON_M9N_GPS_CIRCUIT	Rev A
Date:	Friday, November 07, 2025	Sheet 8 of 9